

Dietary lipids modulate methylmercury toxicity in Atlantic salmon.

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This experiment aimed to study the molecular toxicity of methylmercury (MeHg) in liver, brain and white muscle of Atlantic salmon fed a diet based on fish oil (FO, high dietary n-3/n-6 ratio) compared to an alternative diet mainly based on vegetable oil (VO, low dietary n-3/n-6 ratio). Juvenile salmon were fed decontaminated diets or the FO and VO diets enriched with 5 mg Hg/kg (added as MeHg) for three months. The dietary lipid composition affected the fatty acid composition in the tissues, especially in liver and white muscle. After 84 days of exposure, the liver accumulated three times as much MeHg as the brain and white muscle. Vitamin C content and heme oxygenase, tubulin alpha (TUBA) and Cpt1 transcriptional levels all showed significant effects of MeHg exposure in the liver. TBARS, α -tocopherol, γ -tocopherol, and the transcriptional levels of thioredoxin, heme oxygenase, TUBA, PPARB1, D5D and D6D showed an effect of dietary lipid composition in liver tissue. Effects of dietary lipids were observed in brain tissue for MT-A, HIF1, Bcl-X and TUBA. Interaction effects between MeHg exposure and dietary lipid composition were observed in all tissues. Our data suggest that dietary fats have modulating effects on MeHg toxicity in Atlantic salmon. Copyright © 2011 Elsevier Ltd. All rights reserved.